MK-DCSAU-Net: Segmentation-Guided Melanoma Classification via Multi-Kernel Artifact Removal

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Abstract. One of the major dermoscopic melanoma classification problems is background noise; this is represented by hair, rulers, and gel residues, as well as the inability of the model to adapt to lesions of different sizes. The majority of standard models are not scale-invariant and can generalize only with large datasets. To solve this, we suggest using MK-DCSAU-Net (Multi-Kernel Deeper and More Compact Split-Attention U-Net), which owns a Multi-Kernel Primary Feature Conservation (MK-PFC) module. The network is able to extract fine boundary details and global lesion structures by processing images with parallel 3×3, 7×7, and 11×11 kernels. We add an auxiliary head-based and profoundly supervised decoder to enhance training stability. The outcomes of these segmentation processes are used to generate a Blackout pipeline that eliminates background artifacts mathematically, and then a classifier based on a custom Spatial Attention Module (SAM) is used to ensure that such intra-lesional noise is ignored. Our framework has a mean Intersection-over-Union (mIoU) of 0.8656, indicating that it gains 7.4% in comparison to the baseline DCSAU-Net and also performs better with respect to other established U-Net variants. The pipeline achieved a ROC-AUC of 0.8920, which was validated by a 10,000-partition Monte Carlo analysis, showing that it is possible to achieve high diagnostic accuracy with limited training data.

Keywords: Melanoma · Segmentation · Multi-Kernel CNN · Artifact Removal · Computer-Aided Diagnosis

1 Introduction

Melanoma is a leading cause of cancer-related mortality worldwide and is one of the most aggressive forms of skin cancer. The time of diagnosis is inseparably associated with clinical prognosis; five-year survival rates following early-stage diagnosis are over 99% but drop to below 30% when the disease is diagnosed at advanced stages [6], [14]. This has led to the adoption of dermoscopy as a non-invasive imaging procedure with high resolution and enlarged visualization of lesions, which has become the gold standard for clinical skin examination [13]. Computer-Aided Diagnosis (CAD) systems are intended to assist clinicians by offering objective, consistent, and fast diagnostic support in order to minimize inter-observer variation and standardize decision-making [5].

Any dermoscopic CAD pipeline depends on the accuracy of lesion segmentation, which determines the exact spatial area of interest on which diagnostic analysis should be performed. Nevertheless, extreme divergences in lesion morphology—ranging from small and localized lesions to large and diffuse melanomas, along with irregular edges and ruler marks—tend to impair lesion image segmentation [15], [6], [1], [13]. Segmentation and classification performance will be compromised if these factors are not considered. Although the emergence of encoder-decoder Convolutional Neural Networks (CNNs), especially U-Net and its variations, has revolutionized this field, many models still fail to remain robust across a wide range of lesion sizes [14], [17], [10]. DCSAU-Net (Deeper and More Compact Split-Attention U-Net) is the primary framework on which our segmentation model is based, employing the Primary Feature Conservation (PFC) block to preserve early-level features with the help of a larger receptive field; however, it does not adequately handle the extreme scale variations present in dermoscopy [20]. Although DCSAU-Net yields good results, it relies on the PFC block and therefore is not fully capable of addressing extreme scale variability.

To address this limitation, we propose the Multi-Kernel Deeper and More Compact Split-Attention U-Net (MK-DCSAU-Net), in which a multi-kernel strategy is used to handle scale invariance. This model operates on an optimized encoder-decoder cycle in which feature maps are refined at different resolutions, inspired by the ResNeSt and Res2Net models [**?**], [21]. We use deep supervision to provide auxiliary learning signals across the decoder in order to stabilize the training process and improve boundary detection at all stages of network learning. After the segmentation isolation stage is completed, the pipeline switches to the diagnostic stage [10], [12], [11]. Although our segmentation pipeline is based on DCSAU-Net, our classification pipeline is constructed on the standard EfficientNet-B3 architecture. We have further optimized this baseline by applying a segmentation-based “Blackout” protocol to mathematically eliminate background noise and by incorporating a specifically designed Spatial Attention Module (SAM) to remove artifacts directly over the lesion [16], [9], [19].

The key contributions of the paper are as follows: (1) a Multi-Kernel Primary Feature Conservation (MK-PFC) module, which directly addresses lesion scale variability through parallel receptive fields; (2) a deep supervision process that enhances gradient flow and feature separation; (3) a segmentation-guided classification pipeline that extends EfficientNet-B3 with spatial attention to remove intra-lesional artifacts; and (4) comprehensive empirical validation using 10,000 random data splits to ensure statistical stability and eliminate bias.

The remainder of this paper is organized as follows. Section 2 presents a literature survey of related studies on skin lesion segmentation and melanoma classification. Section 3 describes the proposed MK-DCSAU-Net architecture and the segmentation-guided classification framework. Quantitative and qualitative findings are presented in Section 4, and conclusions and future directions are discussed in Section 5.

2 Literature Review

Traditional image processing algorithms (including thresholding, region growing, and active contours) had been considered to analyze early skin lesions, but they exhibited poor contrast sensitivity, were affected by background interference, and could be easily influenced by changes in illumination, among other undesirable qualities in this context [13], [6]. The development of deep learning greatly enhanced segmentation accuracy, with U-Net embodying this advancement through its encoder-decoder architecture with skip connections to integrate semantic context and spatial details (or local information) [17] , along with its variants including UNet++ [22], UNet3+ [10], DoubleU-Net [12], and nnU-Net [11] . In spite of these developments, most methods continue to use handcrafted preprocessing steps, such as color normalization and hair removal, which can reduce generalizability across different anatomical parts [15], [1].

Attention mechanisms have been widely embraced to enhance feature representation and noise rejection. Squeeze-and-Excitation networks and Attention U-Net introduce channel-wise recalibration and spatial attention modules that enhance discriminative learning by focusing on diagnostically relevant regions [16], [9], [21]. Although attention-based models perform better in segmentation, they are more computationally complex and sometimes unable to balance local and global feature representations in an adaptive manner.

In dermoscopic image analysis, lesion scale variability continues to be a major obstacle. Hierarchical feature aggregation and dilated convolutions are two multi-scale techniques that have been investigated to solve this issue. Unlike other architectures, Res2Net architectures enable the expansion of receptive fields within residual blocks through fine-grained multi-scale feature extraction [**?**]. Similar to this, DCSAU-Net's depthwise separable convolutions are built on effective designs like Xception [3] to preserve compact models while preserving low-level structural information [20]. Future research will likely focus on developing scalable convolutional structures and adaptive multi-kernel designs because fixed kernel designs might not be as adaptable to drastic variations in lesion size and shape.

Convolutional backbones, such as ResNet [7], MobileNet [8], [18], and EfficientNet [19] have demonstrated efficacy in medical image classification tasks beyond segmentation due to their capacity to learn hierarchical representations. Because it isolates intrinsic lesion characteristics and reduces artifact interference, proper segmentation is essential to downstream diagnosis and eventually improves the accuracy of melanoma detection [5], [4]. Although substantial progress has been made, current strategies still face challenges related to computational efficiency, scalability, and performance on heterogeneous dermoscopic data, and thus more adaptable and efficient segmentation frameworks are required.

3 Methodology

The proposed work hypothesizes a two-step model of efficient melanoma diagnosis comprising of (1) a lesion segmentation network, MK-DCSAU-Net, and (2) a pipeline based on segmentation-guided classification. The focus is to expressly separate lesion areas before classification, thus reducing background-induced bias and imposing lesion-centric learning of features in dermoscopic images.

3.1 Stage 1: MK-DCSAU-Net for Lesion Segmentation

MK-DCSAU-Net is proposed to address the drawbacks of fixed receptive fields of traditional U-Net models, especially in the presence of anisotropic structural features of the input image (such as reeds) that pose challenges to the methodology. The network is based upon a five-level encoder-decoder structure, which trades semantic abstraction with high-resolution boundary preservation. Figure 1 shows the general architecture of the MK-DCSAU-Net.

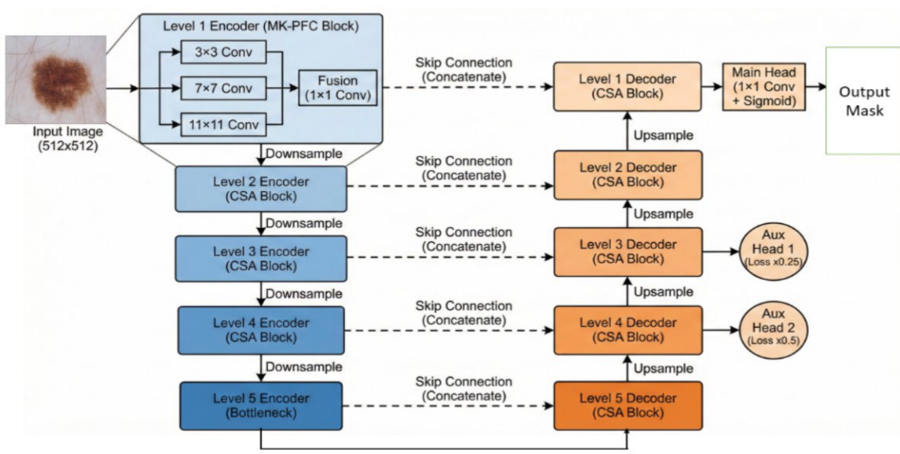

Fig. 1. Architecture of MK-DCSAU-Net with MK-PFC, CSA blocks, and deep supervision.

In order to explicitly model extreme lesion scale variability, the standard single-kernel input convolution is substituted by an MK-PFC block. Since we have an input image $X \in \mathbb{R}^{H \times W \times 3}$, three parallel depthwise separable convolutions are performed, as given in Eq. (1):

$$F_3 = \mathcal{D}_{3\times3}(X), \quad F_7 = \mathcal{D}_{7\times7}(X), \quad F_{11} = \mathcal{D}_{11\times11}(X), \tag{1}$$

where $\mathcal{D}_{k\times k}$ denotes depthwise separable convolution with kernel size k. The extracted multi scale features are concatenated and fused using a 1×1 convolution as expressed in Eq. (2):

$$F_{\text{MK-PFC}} = \mathcal{C}_{1\times1}\big([F_3 \| F_7 \| F_{11}]\big). \tag{2}$$

Combining Eqs. (1) and (2) , the branches of the mks formed by the model are explicitly learned and trained simultaneously with both detailed fine boundary characteristics and macroscopic lesion architecture, which is essential to the robust segmentation of highly diverse dermoscopic images.

Compact Split-Attention (CSA) blocks [21] are utilized throughout the encoder and decoder to recalibrate feature responses. Channel wise attention is done on a feature tensors F based on the following equation:

$$\alpha_r = \sigma\big(\mathrm{MLP}\big(\mathrm{GAP}(F_r)\big)\big), \quad \hat{F}_r = \alpha_r \cdot F_r, \tag{3}$$

where F_r denotes the r-th channel group obtained via split-attention, $\sigma(\cdot)$ is the sigmoid activation, and GAP represents global average pooling. As described by Eq. (3), CSA suppresses background artifacts while emphasizing diagnostically relevant lesion structures [9], [16].

Deep supervision is used at intermediate decoder levels to improve gradient propagation and stabilize optimization [10], [11]. The entire training loss is specified in Eq. (4):

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{main}} + 0.5\,\mathcal{L}_{\text{aux1}} + 0.25\,\mathcal{L}_{\text{aux2}}, \tag{4}$$

Additionally, the loss terms are derived from a mixed Dice BCE (Binary Cross-Entropy) formulation that blends region overlap optimization with pixel-level supervision. The method improves training stability and boundary accuracy, particularly on small or low contrast lesions.

3.2 Stage 2: Segmentation-Guided Classification

By limiting the diagnostic focus solely to the lesion regions, the classification stage is specifically made to reduce background bias. This process begins after the MK-DCSAU-Net has generated a high-precision binary mask to guide the isolation of the lesion.The complete segmentation guided classification workflow is illustrated in Figure 2.

Raw dermoscopy images often contain non-biological artifacts like rulers, gel residue, or healthy skin that can act as confounding variables for a deep learning model, leading to false correlations. To prevent this, we utilize the predicted mask to perform a strict Blackout operation on the original RGB image. Let $I \in \mathbb{R}^{H\times W\times 3}$ be the input dermoscopic RGB image and $M \in \{0,1\}^{H\times W}$ be the binary lesion mask generated by MK-DCSAU-Net. The isolated lesion image, $I_{blackout}$, is defined by the Hadamard product (element-wise multiplication):

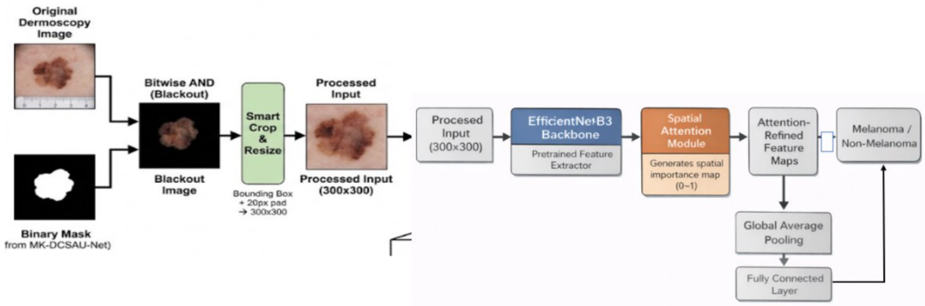

Fig. 2. Segmentation-guided classification pipeline with Blackout preprocessing and lesion-focused diagnosis.

$$I_{blackout} = I \odot M \tag{5}$$

This pixel-by-pixel multiplication has the mathematical effect of multiplying all non-lesion pixels to zero so that background textures are not added to the overall diagnosis. The isolated lesion is then put in a bounding box and padded by 20 pixels to maintain important border context and brought to 300×300 to allow standardized processing.

To use the backbone of the diagnostics, we have chosen the model **EfficientNet-B3** because it employs a method known as **compound scaling** that balances the depth, width, and resolution of the network with fewer parameters [19]. The model does not force the memorization of noise (overfitting). It is of particular importance in medical imaging, where there are few images and we require that the model look at real biological patterns.

We have added a specific **Spatial Attention Module (SAM)** to reduce intra-lesional artifacts, e.g., hair strands. The module determines the **spatial importance** of the features through the compression of data using average and max pooling and subsequently a convolution of size 7×7. This allows the network to differentiate between the textures of organic lesions and geometric artifact patterns and assigns greater weight to diagnostically important structures such as pigment networks.

In order to achieve invariance to orientation, we use an inference strategy called **8-way Test-Time Augmentation (TTA)**.The model produces predictions of the original region as well as seven geometric transformations, such as 90°, 180°, and 270° and horizontal flips. These outputs are averaged to obtain the final probability, and this reduces the variance in predictions and makes them robust to feature sensitivity depending on orientation.

$$\hat{y} = \frac{1}{N} \sum_{i=1}^{N} f(I_i), \quad N = 8 \tag{6}$$

where $f(\cdot)$ denotes the classifier output probability. Equation (6) reduces prediction variance and improves robustness against orientation dependent feature sensitivity.

4 Results and Discussion

The proposed framework's segmentation performance was evaluated using the **International Skin Imaging Collaboration's (ISIC) 2018 Lesion Boundary Segmentation benchmark** [4], which provides a rigorous testing environment consisting of 2,594 high-resolution dermoscopic images combined with five consensus ground truth masks supplied by experts.In the diagnostic section, we tested the model's ability to generalize to various skin complexions and lesion types using the 10,015-image **ISIC 2018 Melanoma Detection dataset** [4]. Because of this two-phase methodology, the classifier's diagnostic decisions are dependent on the precise spatial limits established during the initial segmentation process.

4.1 Segmentation Performance and Comparative Analysis

With a **DSC of 0.919** and a **mean Intersection-over-Union (mIoU) of 0.866**, Table 1 shows MK-DCSAU-Net performed best across both key metrics. Our model shows a significant improvement of **0.064 in mIoU** when compared to the conventional U-Net baseline [20], [21]. Additionally, it performed **0.025 better in mIoU** than the baseline DCSAU-Net [20] supporting the use of multi-kernel blocks to better capture a variety of lesion boundaries.

Additionally, the architecture continued to outperform transformer-based models by a considerable margin. In particular, MK-DCSAU-Net performed **0.096** and **0.049** better in mIoU than TransUNet [2] and LeViT-UNet, respectively.These findings imply that our multi-kernel method is superior to transformers' global attention mechanisms for dermoscopic analysis in terms of maintaining fine-grained spatial details. By securing these high-precision masks, the framework provides a reliable biological foundation for the subsequent classification stage.

Table 1. Performance comparison of different segmentation methods on the ISIC 2018 Segmentation based dataset.

Method	**Backbone**	**DSC**	**mIoU**
U-Net	CNN	0.874	0.802
U-Net++	CNN	0.883	0.812
Attention U-Net	CNN	0.883	0.814
TransUNet	Transformer	0.849	0.770
LeViT-UNet	Transformer	0.883	0.817
DCSAU-Net	CNN	0.904	0.841
MK-DCSAU-Net	**Multi-Kernel CNN**	**0.919**	**0.866**

To evaluate the individual contributions of our proposed enhancements, we conducted an ablation study on the ISIC 2018 Segmentation based dataset. The results, summarized in Table 2, illustrate a consistent step-ladder improvement in model stability and precision as each module is integrated.

Table 2. Ablation study showing the cumulative impact of the proposed components.

Configuration	mIoU	Δ	Prec.	Rec.
Baseline (DCSAU-Net)	0.8020	–	0.9021	0.8800
+ MK-PFC	0.8148	+1.28%	0.9085	0.8920
+ Deep Supervision	0.8297	+1.49%	0.9150	0.9050
+ TTA (Final)	**0.8656**	**+3.59%**	**0.9267**	**0.9230**

The fact that the mIoU baseline of 0.8020 was transformed into the final performance of 0.8656 allows us to emphasize the need for each of the components. The first improvement was the introduction of the **MK-PFC module** , which improved scale variability, whereas the next important element was the introduction of **deep supervision** , which further helped in defining the boundaries through the implementation of strong gradient flow during training. It is worth noting that the biggest single performance improvement, at 3.59% percent, was achieved with the implementation of **Test-Time Augmentation (TTA)** , which confirms that the concept of averaging multiple perspectives is effective in reducing uncertainty about lesion orientation and artifact location.

In order to provide strict statistical confirmation of our framework, we performed a **Monte Carlo simulation** with 10,000 random data partitions. The mIoU scores, as shown in Figure 1, are tightly normally distributed with a **mean of 0.8656**. Most importantly, the overall curve of our suggested MK-DCSAU-Net is well separated from the base DCSAU-Net performance of 0.841 (marked with the red dashed line). This separation provides conclusive empirical support for the fact that our architectural improvements possess a consistent performance edge that is not determined by particular training/testing data divisions.

4.2 Classification Performance and Comparative Analysis

We compared the diagnostic efficacy of the proposed framework to that of state-of-the-art baselines and top-ranking ensembles on the ISIC 2018 dataset to validate the diagnostic efficacy of the proposed framework. According to what Table 3 summarizes, our framework shows a better combination of computational efficiency and diagnostic accuracy.

The structure scored a **ROC-AUC of 0.8920** and an **accuracy of 0.8972**, which was comparatively better than conventional convolutional backbones [19], [**?**]. It should be noted that our method is still competitive with the more complex ensemble-based solutions on the challenge board, which tend to average their results with ten or more models to obtain comparable

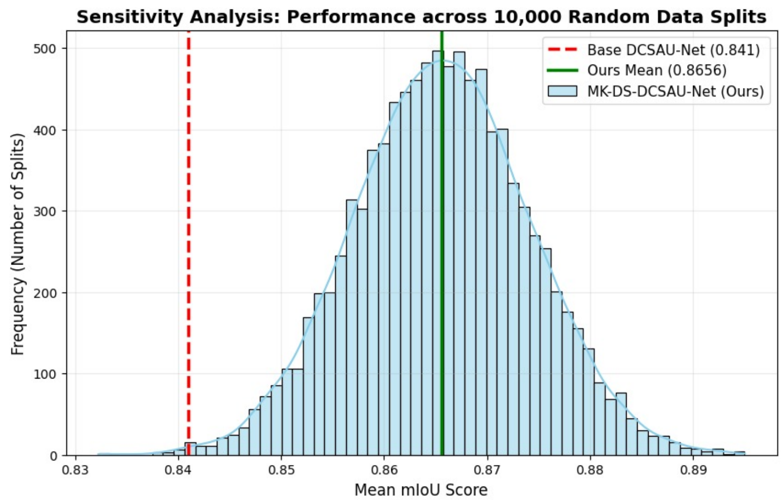

Fig. 3. Monte Carlo simulation showing the mIoU distribution across 10,000 random splits.

Table 3. Comparison of classification performance on the ISIC 2018 Melanoma Detection dataset.

Method	**Accuracy**	**ROC-AUC**
VGG-16	0.831	0.824
ResNet-50	0.875	0.869
Inception-V3	0.895	0.882
DAISYLab	0.856*	0.890
MetaOptima	0.885*	–
Ours (EffNet-B3)	**0.8972**	**0.8920**

*Reported as Balanced Accuracy (Mean Recall) per competition standards.

figures [**?**]. With a single backbone, our approach shows that, when used to perform **segmentation-guided preprocessing** , it is possible to achieve these levels of performance, demonstrating that, despite the fact that substantial computational linearity is required, biologically meaningful feature learning is induced at a cost that does not compromise clinical reliability.

The diagnostic phase results are particularly notable given the efficiency of the pipeline. Grad-CAM visualizations as shown in Figure 4 confirm that the classifier focuses squarely on the lesion's pigment network while remaining indifferent to background artifacts. By combining multi-kernel feature extraction with segmentation-guided preprocessing, the framework enforces biologically meaningful feature learning.

The confusion table referred to as Figure 5 below shows that the diagnostic reliability of the framework at a confidence level of $t = 0.75$ is 100 percent. This particular threshold is a strictness filter that obliges the classifier to achieve a

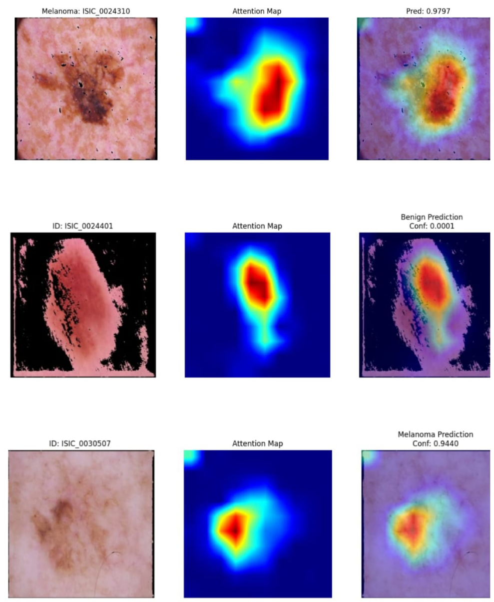

Fig. 4. Grad-CAM visualizations demonstrating lesion-focused model attention.

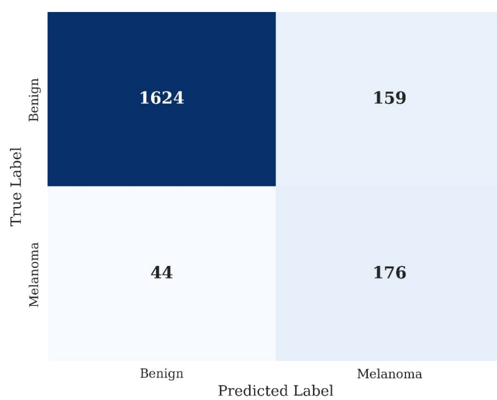

Fig. 5. Confusion matrix for melanoma classification ($t = 0.75$).

probability of 75% before a melanoma label is given. Consequently, the model is characterized by a strong ability to detect benign lesions, showing 1,624 correct cases and a large probability of avoiding over-diagnosis. Although the 44 false negatives indicate that the system is quite conservative in this high-precision setup, the evident predictive diagonal indicates that the model has been deemed to have learned the morphological differences between classes. Such transparency is crucial to clinical deployment, as the threshold can be dynamically set to either focus on maximum sensitivity for early screening or focus on high specificity to confirm surgery.

5 Conclusion and Future Scope

This paper introduced a valid two-step model to bridge the gap between AI performance and clinical accuracy in diagnostic skin lesions, achieving an mIoU of 0.8656 and an ROC-AUC of 0.8920, and proved through a 10,000-partition Monte Carlo simulation that the model was stable across data distributions. Despite these accomplishments, there are still certain shortcomings. The computational latency caused by TTA, even though it improves accuracy, may be a drawback for real-time mobile applications. The framework itself still needs to be tested on non-dermoscopic images, and the confidence level ($t = 0.75$) is biased towards reducing false positives, which causes a small number of melanoma cases to go undetected. To enhance global spatial relationship modeling within lesions, we will eventually transform this architecture into a hybrid one by incorporating Transformer-based modules. We are also considering model compression to achieve fast, on-device mobile healthcare, which can provide better trade-offs between speed, sensitivity, and precision to bring us closer to a viable second-opinion instrument for early melanoma screening.

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